

business and residential customers. Passive optical networks use optical fibre and optical power splitters to connect the OLT at the local exchange to the ONU at subscriber's premises (Fig. 5). No electrical or electronic components are used between these points, all being passive components.

This approach greatly simplifies network operation and maintenance, and reduces the cost. PON, being a point-to-multipoint architecture, requires much lesser fibres as compared to point-to-point topologies. The key network elements of a PON system are OLT, ONU, optical splitter and network management system (NMS).

OLT resides in the telecom service provider's central office and provides traffic aggregation and switching functionality between the core network and PON interfaces. OLT devices support management functions and can manage up to 128 downstream links. In practice, it is common for only 8 to 32 ports to be linked to a single OLT in the central office. Thus, one OLT serves multiple ONUs/ONTs.

The access node installed within user premises for network termination is termed as ONT. Access node installed at any other location, such as curb/cabinet/building, is known as ONU. ONU/ONT provides access to

the users and interfaces to uplink local traffic towards OLT.

Passive optical splitters/combiners provide connectivity between OLT and several ONUs/ONTs. These splitters (distributed or single staged) are available in several options like 1:4, 1:8, 1:16, 1:32 and 1:64.

Network management functions are performed from OLT through the NMS. Time division multiplexed (TDM) data is sent downstream from the OLT towards each ONU where the appropriate portion is extracted for local use. In the upstream direction, a time domain multiple access (TDMA) protocol allocates slots for data transmitted from each ONU to communicate back to the OLT without any contention between different subscribers. Typical distance between OLT and ONU can be up to 35 kms.

Based on Ethernet technology to create a passive optical infrastructure, PON is classified as Ethernet-PON (EPON) and gigabit Ethernet-PON (GPON). The key features of various PONs are summarised in Table I.

PON requires fewer fibres because subscribers are connected via dedicated distribution fibres to the site, and they share the optical distribution network (ODN) trunk fibre back to the central office. Fig. 6 shows the lesser fibre requirement for PON (EPON and GPON) as compared to the topologies of point-to-point (P2P) and point-to-multipoint (P2M) switched Ethernet.

Point-to-point Ethernet might use either N or 2N fibres, and would have 2N optical transceivers. Point-to-multipoint switched Ethernet uses one or two trunk fibres and thus would save fibres and space in the telecom service provider's central office (CO). But it would use 2N+2 optical transceivers and would require electrical power in the field.

PON also uses only one trunk fibre and thus minimises fibres and space in the CO, and it also uses only N+1 optical transceivers. Most importantly, it requires no electrical power in the field. The drop throughput can be up to the line rate on the trunk link. EPON can

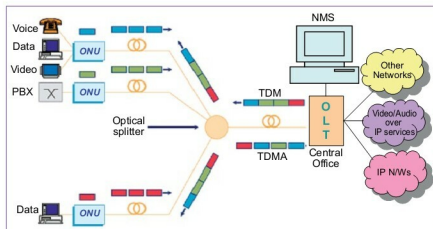


Fig. 5: Passive optical network architecture

Key Features of Various PON Standards

Features	BPON	EPON	GPON
Standard	FSAN & ITU-T SG15 (G-983 series)	IEEE 802.3ah	FSAN & ITU-T SG15 (G-984 series)
Download	622Mbps	1.25Gbps	2.5Gbps
Upload	155.52Mbps	1.25Gbps	2.5Gbps
Downstream wavelength	1490nm & 1550nm	1490nm	1490nm and 1550nm
Upstream wavelength	1310nm	1310nm	1310nm
Layer-2 protocols	ATM	Ethernet	ATM, Ethernet, TDM over GEM
Frame	ATM	Ethernet	GPON encapsulation method
Max. distance (OLT to ONU)	20km	10km and 20km	20km (supports logical reach up to 60 Km)
Split ratio	1:16, 1:32 and 1:64	1:16 and 1:32	1:16, 1:32 and 1:64
Downstream security	AES: Advanced encryption standard 128-bit	Not defined	AES: Advanced encryption standard (counter mode)
Forward error correction	None	Yes	Yes
Protection switching	Support multiple protection configuration	None	Support multiple protection configuration